

3. Exogenetic Movements Part-1

Std 9th

Sub – Geography

Q 1. Answer in brief.

(a) What is mechanical weathering ?

Ans. 1. The disintegration of rocks without any changes in its chemical composition is called as mechanical weathering.

2. The minerals in the rocks expand due to heat and contract when the temperature decreases.

3. Due to such contracting and expanding tension, develops in the rock.

4. Each mineral reacts differently to the heat some may expand more while some do not expand much.

5. Consequently, the tension in the rock increases and decreases which leads to the formation of cracks in the rock and they break.

(b) What are the main types of chemical weathering?

Ans. The decomposition of rocks due to the changes in its chemical composition is called as chemical weathering.

The main types of chemical composition are as follows:-

a. Carbonation

1. When the rain water mixes with the carbon dioxide in the atmosphere it leads to the formation of carbonate acid.

2. Many rocks like limestone gets easily dissolved in such a acids.

b. Solution

1. Some minerals in the rocks get dissolve in the water.

2. Because Of this solution, alkalines dissolve in the water and make them brittle.

c. Oxidation

1. This process occurs in the rocks which have iron present in them. The iron comes in contact with the water and a chemical reaction takes place between the iron and oxygen.

2. Hence, a reddish layer develops on the surface of the rock. This is called as rust.

3. It occurs in the regions of high rainfall.

(c) How does biological weathering occur?

Ans. 1. It is the process in which the rocks are broken down into small fragments and fine particles due to the activities of plants, animals and human beings.

2. Roots of trees enter the joint and cracks of the rocks in such of moisture.

3. When the roots grow bigger, they create tension in the rocks and start breaking them.

4. Animals like mice, rats and rabbits dig holes anthills etc. which weakens the rocks and makes them loose and break them.

5. Some algae and other flora grow in the rocks. They also help in weathering.

6. Hence, the weathering caused by living organisms is called as biological weathering.

(d) Distinguish between weathering and mass wasting.

Ans.

weathering	mass wasting
1. Breaking of weakening of rocks is called as weathering.	1. When the materials of weathered rocks come down because of gravity and accumulate near the foothills and gentle slope. It is called as mass wasting.
2. Weathering can either be natural or by living organisms.	2. Mass wasting can occur only naturally .
3. Weathering is of three types biological, chemical and mechanical.	3. Mass wasting is of two types slow and fast.

Q 2. Write whether the statements are true or false. Correct the incorrect ones.

(a) Climate affects earthquakes.

Ans. False – Internal movement affects earthquakes

(b) Mechanical weathering is less effective in humid climates.

Ans. True

(c) Mechanical weathering happens on a large scale in dry climates.

Ans. True

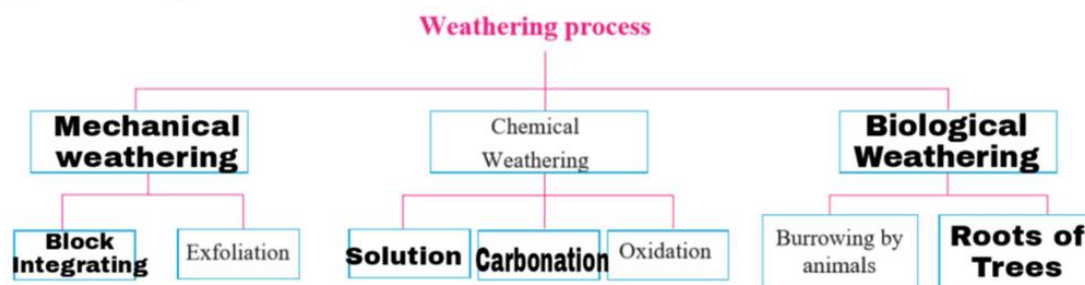
(d) The breaking down of rocks into smaller particles is called weathering.

Ans. True

(e) Lateritic rocks are formed through exfoliation.

Ans. False - Lateritic rocks are formed through oxidation.

Q 3. Complete the flowchart below.



Q 4. Identify the type of weathering from the given description.

(a) Some animals live inside the grounds by making burrows.

Ans. Biological weathering

(b) The rock rusts.

Ans. Chemical weathering

(c) Water which has accumulated in the crevices of the rocks freezes. Consequently, the rock breaks.

Ans. Mechanical weathering

(d) The pipes supplying water in colder regions break.

Ans. Mechanical weathering

(e) Sand formation occurs in deserts

Ans. Mechanical weathering